

EDUCATION

Master of Science in Computer Software Engineering, Northeastern University, Boston, MA (GPA: 3.8/4.0)
Relevant Courses: Generative AI & LLMs, Machine Learning, Algorithms & Data Structures, Object-Oriented Design, Web Development, Software QA

Bachelor of Engineering in Information Technology, LDRP ITR, Gandhinagar, India (GPA: 9.1/10)

Sep 2024 - May 2026

Sep 2020 - May 2024

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C#, SQL, C++, PHP

AI & ML: LLMs, RAG, LangChain, FAISS, Pinecone, Chroma, Fine-Tuning, LoRA, HuggingFace, PyTorch, TensorFlow, Scikit-learn, CNNs

Backend: Django, FastAPI, Node.js, Express.js, Spring Boot, .NET, Microservices, Celery, Kafka

APIs & Security: REST APIs, gRPC, Protocol Buffers, JWT, OAuth, RBAC

Frontend: HTML5, CSS3, React, Angular, Redux, Tailwind CSS, Responsive Design, PWA, WCAG Accessibility, Figma

Databases: MySQL, PostgreSQL, Microsoft SQL Server, Oracle SQL, MongoDB, Redis, Query Optimization, Indexing, Caching

Cloud & DevOps: AWS (EC2, S3, SQS), Docker, Kubernetes, Nginx, GitHub Actions CI/CD, Jenkins, Azure DevOps, Linux, Git

WORK EXPERIENCE

Software Engineer Intern, R K Farmer's Care, India
• Built a web portal and Android app serving 1,000+ rural farmers on a shared Python/PHP backend, a single MySQL schema covering marketplace, orders and payments and a shared CNN inference endpoint
• Wrote the REST APIs in Python and PHP. Added JWT auth and role-based access so admin and farmer permissions stayed separate
• Added AgroExpert, CNN crop and disease detection from a photo upload, 20-30% more accurate than checking by eye
• Shipped a multilingual interface with weather data, subsidy data and an NLP chatbot so farmers could use the app in their own language

Artificial Intelligence Intern, Code Clause, Pune, India
• Built a Dockerized ML pipeline for data preprocessing and model training that cut environment setup time for the team by about 40%
• Fine-tuned CNN models for a healthcare prediction task and improved performance 15-20% on precision, recall and F1, with validation checks on the image data that reduced failed training runs

Software Engineer Intern, Dreams Technology, Gandhinagar, India
• Shipped 10+ responsive pages for search, booking, payment and the admin in HTML5, CSS3 and JavaScript with cross-browser support
• Built REST APIs in Python and Express.js, secured with HTTPS. Set up SAML single sign-on through Azure Active Directory for 50+ users
• Wrote unit and integration tests in PyTest and Selenium that caught defects before release. Shipped changes through Agile CI/CD

Dec 2023 - Jul 2024

Oct 2023 - Nov 2023

May 2023 - Jun 2023

PROJECTS

Agentic AI System for Distributed NL-to-SQL

- Split an NL-to-SQL system into two services over gRPC with Protocol Buffer contracts. One reads the question, the other writes the SQL
- Used FAISS to pull the real schema before generating a query. This cut wrong table and column names compared to a single-model setup
- Put OpenAI and Vertex AI behind one interface so either could be swapped in without changing agent logic, with a Gradio query front end

HuskyTracker, Campus Event Platform

- Built an event platform for students and organizers in React, TypeScript, Node.js and Express with JWT auth, role-based routes, search, RSVP and real-time notifications
- Handled RSVP, waitlist and capacity updates so two users could not claim the same seat. Added MongoDB indexes to keep search fast
- Containerized with Docker and deployed on AWS EC2 behind Nginx. Set up a GitHub Actions CI/CD pipeline for a 5-person team

Fine-Tuned LLMs for Text-to-SQL

- Fine-tuned FLAN-T5 and LoRA-adapted LLaMA-2 on the Spider benchmark. Both beat prompt-only approaches on Exact Match and BLEU
- Built a FAISS RAG pipeline that injects table and column metadata into the prompt, with a Gradio demo returning SQL and SQLite results

Iris Event Management System

- Designed the MySQL schema and backend services in Java and Spring Boot for a 5-role booking system, using Factory and DAO patterns
- Used multithreading so two customers could not book the same venue slot. Built full CRUD for events, venues and bookings

ACHIEVEMENTS

1st Place, MassDOT Billboard Design Contest (Statewide Winner), Massachusetts Department of Transportation, CIM, MAAPA

Winner, Reducing Children's Anxiety in Healthcare Settings, Design Research Week, Northeastern University

May 2025

Mar 2025

ADDITIONAL EXPERIENCE

RSO Proctor (On-Campus Part-Time), Northeastern University, Boston, MA
• Managed access control and visitor authentication systems, flagging anomalies and coordinating with campus security on incidents

Dec 2025 - Apr 2026